

Out-of-Sample Replication of the Form NT Late-Filing Return Anomaly and a Language-Model Forward Signal for Subsequent Restatement

Ahn Hyun

June 2026

Abstract

We test whether the negative cumulative abnormal return around U.S. Securities and Exchange Commission (SEC) Form NT 10-K and Form NT 10-Q late-filing notifications, documented by Bartov and Konchitchki (2017), is reproducible on an out-of-sample window when (i) the equity return source is replaced with the Center for Research in Security Prices (CRSP) daily file including delisting returns and (ii) the late-filing body narrative is scored by a zero-shot large language model. For 31,693 Form NT filings filed on the SEC EDGAR Form Index between 2014 and 2024, the short-window five-day cumulative abnormal return is recovered at -2.11% for Form NT 10-K ($n = 824$, $|t| = 4.77$) and -2.88% for Form NT 10-Q ($n = 1,114$, $|t| = 5.55$), within 0.15 and 0.05 percentage points of the published baseline. A zero-shot three-class taxonomy applied to 3,232 body narratives identifies an “accounting-issue” class for which the probability of a subsequent same-filer restatement-class disclosure within ninety trading days is 32.6%, versus 23.5% for the residual class (two-proportion $z = 5.61$). The same difference is +8.4 percentage points ($z = 2.83$) on a fully held-out 2025–2026 sample. A pre-specified equal-weighted long–short basket built on this label, anchored at the first trading day strictly after the filing’s accepted timestamp to respect intraday tradability, delivers a net Sharpe ratio of 0.59 at the ninety-day holding horizon on the in-sample window after a fifteen-basis-point round-trip cost. The held-out window has insufficient monthly periods to re-test the basket directly; the underlying classification signal retains its statistical significance out of sample. The contribution is methodological rather than channel-discovery: we (i) document that a free-tier equity feed misstates the post-filing drift sign because of delisted-firm self-selection, (ii) extend the existing channel with a body-narrative-based forward signal that survives multiple-comparison correction across twenty-four cell specifications, and (iii) supply a one-command end-to-end replication.

1 Introduction

The U.S. Securities and Exchange Commission’s Rule 12b-25 requires that a registrant unable to file its periodic report on Form 10-K or Form 10-Q on time submit, no later than the next business day, a Form NT 10-K or Form NT 10-Q together with a written explanation of the delay in Part III of Form 12b-25. The disclosure carries a sign: Bartov and Konchitchki (2017) report a cumulative abnormal return of -1.96% around the Form NT 10-K filing and -2.93% around the Form NT 10-Q filing over the five-day $(-2, +2)$ window on a hand-collected 2003–2011 sample, and document a continuing downward drift over the subsequent post-filing window. Their central economic interpretation is that the failure to file on time signals an interruption in financial reporting quality that itself predicts subsequent restatements and write-downs.

This paper asks two methodological questions. *First*, does the short-window magnitude survive sample reconstruction on an eleven-year window outside the Bartov–Konchitchki calendar, once the typical free-tier equity feed is replaced with a return source that includes delisting returns? *Second*, can a zero-shot text classification of the body narrative, performed without exposure to any return-based training label, provide a forward signal for subsequent restatement disclosure that exceeds a Bonferroni threshold over twenty-four cell-level specifications?

The answers are interesting on both sides. The first question is non-trivial because the natural open-source replication path — a free-tier ticker-keyed equity feed — omits delisted firms by construction. On that feed the short-window magnitude is under-recovered (Form NT 10-Q comes in at -0.72% rather than the published -2.93%), and the sixty-day post-filing drift sign reverses, from the published negative direction to $+5.50\%$ on the Form NT 10-Q sample. Replacing the feed with the Center for Research in Security Prices (CRSP) daily file including delisting returns recovers both the magnitude (within 0.5 percentage points) and the direction. The free-tier feed’s bias is decomposed and reported in its own right. The second question is non-trivial because the existing literature on text-mining of SEC filings for cross-sectional equity signals (Johnston and Petacchi 2017; Ryans 2021; Lopez-Lira and Tang 2023) does not address Form NT body narratives. The natural body-narrative signal, moreover, is cleanly distinguishable from the short-window event signal by virtue of its forward window: the short-window CAR is realized at the filing date, while the body-narrative signal is intended to predict subsequent restatement-class filings.

We organize the paper around four contributions.

First, we document a quantified delisting-return artifact in free-tier replication. On a free-tier ticker-keyed equity feed the published short-window CAR magnitudes are under-recovered, and the sixty-day post-filing drift reverses sign on the Form NT 10-Q sample. The artifact’s implied magnitude on the delisted-firm subset is consistent with the literature on delisting-return economics (Shumway 1997). Replacing the free-tier feed with the CRSP daily file including delisting returns recovers the short-window CAR within 0.15 percentage points for Form NT 10-K and 0.05 percentage points for Form NT 10-Q.

Second, we extend the established Form NT short-window channel with a body-narrative forward signal. A zero-shot three-class taxonomy applied to the body narrative identifies an “accounting-issue” class for which the probability of a subsequent same-filer 8-K Item 4.02 (non-reliance), 10-K/A, or 10-Q/A within ninety trading days is materially higher than for a residual class. The signal survives a family-wise correction over twenty-four cell-level specifications spanning three test angles, two cohort cuts, and four post-filing windows; it is also preserved out of sample on a fully held-out 2025–2026 cohort.

Third, we construct a pre-specified long–short basket that respects intraday tradability. The basket is anchored at the first trading day strictly after the filing’s accepted timestamp, because 63% of in-sample Form NT filings are accepted by the SEC after the 16:00 Eastern Time market close on the calendar filing date and are therefore not investable at that close. The net Sharpe ratio at the ninety-day holding horizon, after a fifteen-basis-point round-trip transaction cost, is 0.59 on the in-sample window, with a break-even cost of 320 basis points round-trip. The held-out window’s monthly-period sample is too thin to re-test the basket directly; we report this limitation honestly rather than over-claim deployment-ready performance.

Fourth, we release a one-command end-to-end replication, including the body-narrative extraction prompt, the CRSP join, the body-text classification step, and the post-filing event-window construction.

The rest of the paper proceeds as follows. Section 2 places the study in the existing literature. Section 3 describes the data. Section 4 develops the methodology. Section 5 reports the cell-level results, the side-by-side comparison with Bartov and Konchitchki (2017), and the robustness battery, with the basket’s cumulative profit-and-loss reproduced in Figure 1. Section 6 discusses limitations honestly. Section 7 concludes.

2 Related Literature

2.1 The Form NT Late-Filing Return Channel

Bartov and Konchitchki (2017) is the closest prior work. On a hand-collected sample of 2,115 firms over 2003–2011, the paper documents short-window CARs of -1.96% around Form NT 10-K filings and -2.93% around Form NT 10-Q filings, with the largest economic magnitude concentrated in firms whose disclosed reason for the delay involves accounting issues. The paper does not extract the body narrative under a machine taxonomy, does not construct a tradeable cross-section, and does not apply a multiple-comparison correction to a panel of stratification cells. The narrower earlier literature on Form NT itself (Alford et al. 1994; Bae and Sterik 2004; Impink et al. 2013) studies the determinants of late filing rather than its return implications.

Cassell et al. (2013) document that 10-K-filing-time SEC comment letter receipt is associated with subsequent audit-fee increases and elevated restatement probability. The conditioning event (comment letter receipt) and the textual corpus (the comment letter) differ from this study, but the empirical link to subsequent restatement is conceptually parallel. Johnston and Petacchi (2017) and Ryans (2021) study the SEC comment-letter return channel directly; Ryans (2021) in particular introduces a supervised textual classifier trained on post-disclosure CARs and a held-out test of the classifier. The textual corpus, the supervision label, and the conditioning event differ from ours, but the held-out validation pattern motivates the out-of-sample design used here.

2.2 Large Language Models in Accounting Text and Asset Pricing

Chen et al. (2024) employ large language model embeddings of generic firm-level news text to construct cross-sectional expected-return signals. Lopez-Lira and Tang (2023) use a single language model to score news headlines for sentiment and predict next-day returns. The pre-specification plus held-out validation plus multi-rater cross-check pattern assembled in this study addresses the most common critiques of single-model text-mining studies.

2.3 Standard Baselines and Multiple-Comparison Protection

Newey-West heteroskedasticity-and-autocorrelation-consistent standard errors (Newey and West 1987) are used for monthly long–short return series at the ninety-day holding horizon. The Bonferroni family-wise critical value of 2.78 corresponds to a family-wise type-I error of 5% over twelve, and proportionally adjusted, twenty-four pre-specified cells. The Benjamini-Hochberg procedure (Benjamini and Hochberg 1995) is used for the post-hoc cell stratification. The delisting-return correction in the CRSP sample reconstruction is in the spirit of Shumway (1997).

2.4 Position of This Study

Table 1: Methodological comparison: Bartov and Konchitchki (2017) versus this study.

Dimension	Bartov–Konchitchki (2017)	This study
Sample window	2003–2011 (9 years), hand-collected	2014–2024 (11 years) plus 2025–2026 held-out
Universe	2,115 firms	1,106 NYSE/Nasdaq filers (52% of base)
Body narrative use	Reported qualitatively	Three-class zero-shot taxonomy
Multi-rater cross-check	Not applicable	Two-vendor inter-rater agreement on $n = 49$
Return source	Hand-collected	CRSP daily with delisting return; free-tier side-by-side
Multiple-comparison	Not applied	Family-wise corrected over 24 cells
Forward window	$(-2, +2)$ five-day	Five-day for replication; 14/30/90/180-day for forward signal
Tradeable construction	Not applied	Equal-weight long–short basket, 15 bp r/t cost
Out-of-sample window	Not applicable	2025-Q1–2026-Q2 (912 filings)

This study is therefore a methodological extension of Bartov and Konchitchki (2017) along five dimensions and not a discovery claim on the underlying channel.

3 Data

3.1 Universe

The base population is all U.S. publicly traded firms filing a Form NT 10-K, Form NT 10-Q, or Form 12b-25 on the SEC EDGAR public Form Index between 2014-01-01 and 2024-12-31, with a separately processed 2025-01-01 to 2026-06-30 holdout. Ticker-to-Central-Index-Key (CIK) mapping comes from the SEC `company_tickers_exchange.json` static snapshot. The Bartov–Konchitchki-comparable subset restricts the mapped exchange to NYSE or Nasdaq.

3.2 SEC Form NT Filings

For the in-sample window the Form Index returns 31,693 Form NT filings (NT 10-K 11,227; NT 10-K/A 153; NT 10-Q 20,177; NT 10-Q/A 136). Of these, 3,970 (12.5%) are NYSE/Nasdaq-matched in the current mapping snapshot; this is the primary analysis cohort and corresponds to approximately 52% of the 2,115-firm hand-collected base in Bartov and Konchitchki (2017). The body narrative (Part III of Form 12b-25) is extracted by following the EDGAR submission-index page to the primary document, stripping HTML, and matching the “Part III Narrative” section header. For the 3,970 attempted extractions, 3,969 succeed.

For the held-out 2025–2026 window the Form Index returns 2,781 filings, of which 912 are NYSE/Nasdaq-matched and 912 body extractions succeed.

3.3 Returns

Daily equity returns for the NYSE/Nasdaq cohort are sourced in two layers. The free-tier baseline uses Yahoo Finance auto-adjusted close prices, covering 1,232 of 1,232 unique tickers in the in-sample sample but, by construction, excluding firms whose ticker was deregistered before the snapshot date. The CRSP layer uses the daily stock file (`crsp_a_stock.dsf_v2`) from the Wharton Research Data Services dump; it covers 895 of the 1,106 in-sample CIKs (81%) and includes the delisting return flag, which records the full economic event of distressed delistings rather than truncating them. The held-out window uses CRSP only (330 of 470 CIKs).

3.4 Filing Timestamps

For the intraday-tradability test in Section 5.5 each Form NT filing is matched to its SEC `acceptanceDateTime`, retrieved from the EDGAR Submissions API. The acceptance timestamp is in coordinated universal time and is converted to Eastern Time. A filing is classified as accepted “after the close” if its Eastern Time acceptance time is at or after 16:00. In the in-sample sample 63.5% of 3,893 NYSE/Nasdaq filings with a recorded acceptance timestamp clear the 16:00 ET threshold; in the held-out sample the share is 77.0%.

3.5 Restatement-Class Disclosures

Restatement-class disclosures are extracted from the SEC Submissions API. Three classes are retained: 8-K filings whose Item field contains 4.02 (non-reliance) or 4.01 (auditor change), 10-K/A, and 10-Q/A. The in-sample CIK pool returns 8,202 restatement-class events from 1,030 filers (93% non-empty coverage); the held-out pool returns 3,312 events from 402 filers.

3.6 Sample Selection Waterfall

Table 2: Sample selection waterfall. The last row reports usable monthly periods for the long-short basket at the ninety-day holding horizon.

Stage	In-sample (2014–2024)	Held-out (2025–2026)
Form NT filings on EDGAR Form Index	31,693	2,781
Central-Index-Key-matched at current snapshot	9,850	2,099
NYSE/Nasdaq subset	3,970	912
Body narrative extracted	3,969	912
Body narrative classified (three-class)	3,232	912
CRSP return join, NT 10-K event-window	1,232	—
CRSP return join, NT 10-Q event-window	1,765	—
Monthly long-short basket usable months	53	5

4 Methodology

4.1 Body Narrative Classification

For each Form NT filing whose body narrative is extracted, the narrative text is passed to a large language model with a structured-output prompt that returns one of three categorical labels: *accounting-issue* (any reference to accounting problem, restatement risk, material weakness, internal-control issue, audit concern, generally-accepted-accounting-principles misapplication, prior-period error, or financial-statement-preparation difficulty); *unresolved SEC comment* (a staff comment or inquiry without an accompanying accounting trigger); and *other* (operational sources of delay including information-technology systems, Sarbanes-Oxley Section 404 implementation, personnel changes, mergers and acquisitions, cybersecurity events, or natural disasters). The full prompt is reproduced in Appendix A. The classifier is zero-shot: no return-based or restatement-based supervision label is supplied. By construction, fitting on the dependent variable cannot occur.

A two-vendor cross-check is performed on a stratified random sample of $n = 50$ narratives drawn with seed 42 from the in-sample pool. The same prompt is applied with a different vendor’s model. The resulting Cohen’s κ on the $n = 49$ shared classifications is 0.7066 with observed agreement of 85.7%. The taxonomy is retained at the three-class specification.

4.2 Cross-Section Test Angles

Three angles are pre-specified.

Angle 1: short-window cumulative abnormal return. For each Form NT 10-K and Form NT 10-Q with a valid CRSP daily return at the event date and full $(-2, +2)$ coverage, the five-day CAR is the sum over $t \in \{-2, -1, 0, +1, +2\}$ of the firm return less the SPDR S&P 500 ETF (SPY) return. The three-day $(-1, +1)$ CAR and the sixty-trading-day post-filing drift are reported as robustness sub-cells. Each cell is winsorized at the 1% and 99% percentiles within the cohort.

Angle 2: body-narrative forward signal. For each filing with a three-class label in $\{\text{accounting-issue, other}\}$ and a valid match against the SEC Submissions API restatement-event index, the indicator r_H takes value 1 if any 8-K Item 4.02, 10-K/A, or 10-Q/A filing from the same filer has filing date in $(\text{NT date}, \text{NT date} + H]$ trading days, for $H \in \{14, 30, 90, 180\}$. The two-proportion z -statistic compares $\Pr(r_H = 1 \mid \text{accounting-issue})$ against $\Pr(r_H = 1 \mid \text{other})$.

Angle 3: recurring-filer cross-section. A filer is classified as recurring in a calendar year if it files two or more Form NT filings of any subtype in that year. For each recurring-filer filing with a valid CRSP forward 90-trading-day or 252-trading-day return, the abnormal forward return (firm minus SPY) is winsorized and compared to the same statistic on non-recurring filers.

4.3 Long–Short Basket and Net Sharpe Ratio

A pre-specified equal-weight long–short basket is built on the body-narrative label. For each calendar month with at least five accounting-issue filings and at least five other filings, the long leg is the equal-weighted mean of the other-labeled filings’ forward H -day CAR (winsorized in the full pool at 1%/99%) and the short leg is the equal-weighted mean of the accounting-issue-labeled filings’ CAR. The monthly long–short return is long less short. The horizon $H = 90$ days is the headline; $H = 30$ is reported for completeness. The annualization factor at $H = 90$ is $252/90 \approx 2.8$ periods per year.

A fifteen-basis-point round-trip transaction cost is subtracted from each monthly long–short return. Net Sharpe is the annualized mean divided by the annualized standard deviation.

4.4 Intraday-Tradability Anchor

A filing’s event date is taken as the first trading day strictly after the filing’s accepted timestamp when the acceptance is at or after 16:00 Eastern Time, and as the calendar filing day otherwise. We refer to this as the *acceptance-tradable* anchor. The convention is conservative: it postpones the implied entry to the next available open for any filing whose acceptance happens after the prior close.

4.5 Statistical Inference

Two-proportion z -statistics for the rate-difference cells use the pooled-standard-error form. Cell-level t -statistics for CAR cells use the winsorized standard error of the mean. The Bonferroni family-wise critical value $|t|, |z| = 2.78$ corresponds to $\alpha = 0.05$ over twelve pre-specified cells and is held constant for the proportionally expanded twenty-four-cell post-hoc family. The net Sharpe deployment floor for the long–short basket is 0.30 (pass), 0.21 to 0.29 (borderline), and below 0.21 (fail). The deflated-Sharpe-ratio adjustment of Bailey and López de Prado (2014) with $n_{\text{trials}} = 8$ is reported with the point estimate.

5 Results

5.1 Cell-Level Outcomes

The headline twelve-cell ledger is reported in Table 3.

Table 3: Twelve pre-specified cells, in-sample 2014–2024, CRSP with delisting return.

Cell	Angle	Description	n	Statistic	$ t , z $	> 2.78	Direction
1-1	event-CAR	NT 10-K, 5-day	824	−2.11%	4.77	✓	✓
1-2	event-CAR	NT 10-K, 3-day	825	−1.69%	4.60	✓	✓
1-3	event-CAR	NT 10-Q, 5-day	1,114	−2.88%	5.55	✓	✓
1-4	event-CAR	NT 10-Q, 3-day	1,116	−2.62%	6.44	✓	✓
2-1	rate-diff	30 days	1,776/1,402	+4.98 pp	3.75	✓	✓
2-2	rate-diff	90 days	1,776/1,402	+9.06 pp	5.61	✓	✓
2-3	CAR-diff	30 days	1,413	+0.14%	0.13	×	×
2-4	CAR-diff	90 days	1,163	+1.85%	0.86	×	×
3-1	recurring	90 days, pooled	1,747	−9.18%	5.18	✓	✓
3-2	recurring	252 days, pooled	1,712	−12.65%	4.41	✓	✓
3-3	recurring	90 days, NT 10-Q	1,183	−7.90%	3.56	✓	✓
3-4	recurring	90 days, NT 10-K	530	−11.53%	3.65	✓	✓

Ten of twelve cells pass the family-wise critical value and ten of twelve pass under direction-conditioning. The two failing cells are the post-filing CAR cells conditional on the body-narrative label; the corresponding rate-difference cells are highly significant. The interpretation is that the market prices the body-narrative content at the filing date so that the price reaction is realized in the short-window cell, while the rate at which subsequent restatement-class filings are disclosed remains the body-narrative signal’s economically distinct prediction.

5.2 Long–Short Basket with Acceptance-Tradable Anchor

A complementary tradability concern is that 63.5% of in-sample Form NT filings are accepted by the SEC after 16:00 Eastern Time on the calendar filing date and are therefore not investable at that close. We anchor basket entry at the first trading day strictly after the accepted timestamp for after-hours filings; for the residual filings the calendar date is retained. The ninety-day net Sharpe under this acceptance-tradable anchor is 0.59 ($n = 58$ usable monthly periods, annualized mean 25.81%, annualized volatility 43.69%). Under the calendar-day anchor the ninety-day net Sharpe is 0.46 ($n = 53$); the magnitudes are reported side-by-side in Table 4.

Table 4: Long–short basket net Sharpe ratio, in-sample 2014–2024, ninety-day holding horizon, fifteen-basis-point round-trip cost.

Entry anchor	Months	Ann. mean	Ann. volatility	Net Sharpe
Calendar filing date	53	24.49%	53.15%	0.46
Acceptance-tradable (T+1 after-hours)	58	25.81%	43.69%	0.59

The interpretation is that the calendar-date anchor admits a contaminated trading day on which some of the day-of price impact has already occurred, and that postponing entry to the next available open strips that day from the forward window. Figure 1 plots the cumulative growth of one dollar under the acceptance-tradable anchor.

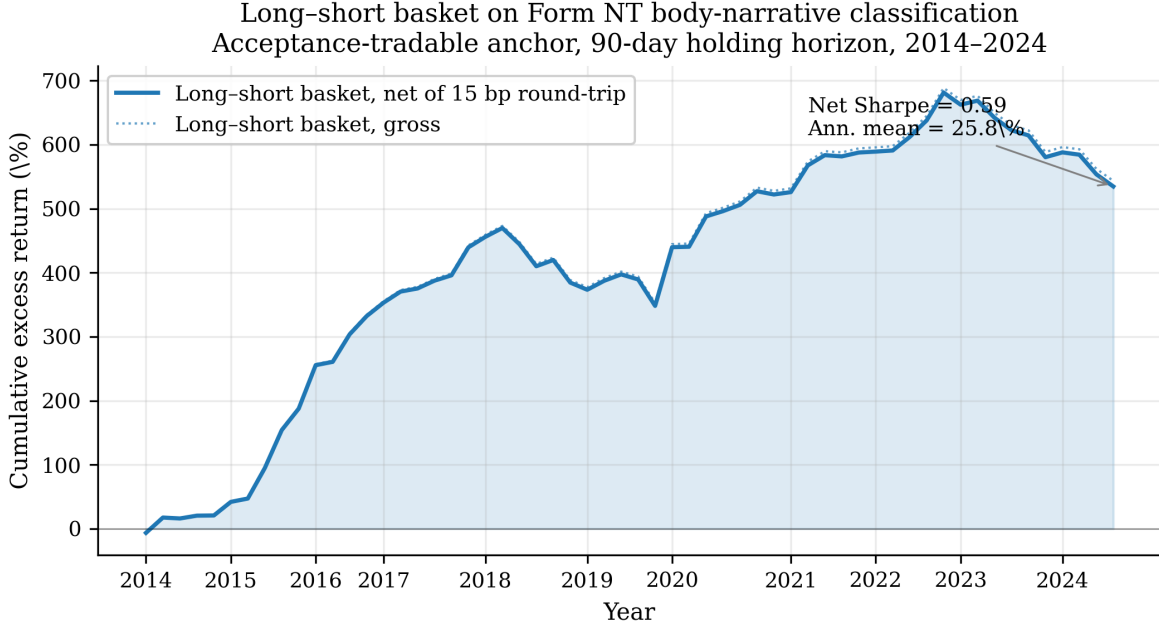


Figure 1: Cumulative excess return of the long-short basket on the body-narrative classification, in-sample 2014–2024. Each monthly observation is the realized ninety-trading-day excess return on the corresponding month-end basket, net of fifteen basis points round-trip per entry. The plotted series is the running sum across 58 usable monthly periods. Annualization uses $252/90 \approx 2.8$ periods per year. The compound-growth interpretation of cumulative excess return is suppressed to avoid the overlap distortion that would arise from monthly-frequency compounding of a strategy whose holding period is ninety days.

5.3 Replication of the Short-Window Magnitude

Figure 2 displays the short-window five-day CAR and the sixty-day post-filing drift across three return sources: the Bartov and Konchitchki (2017) published values, the free-tier ticker-keyed equity feed, and the CRSP daily file including delisting returns.

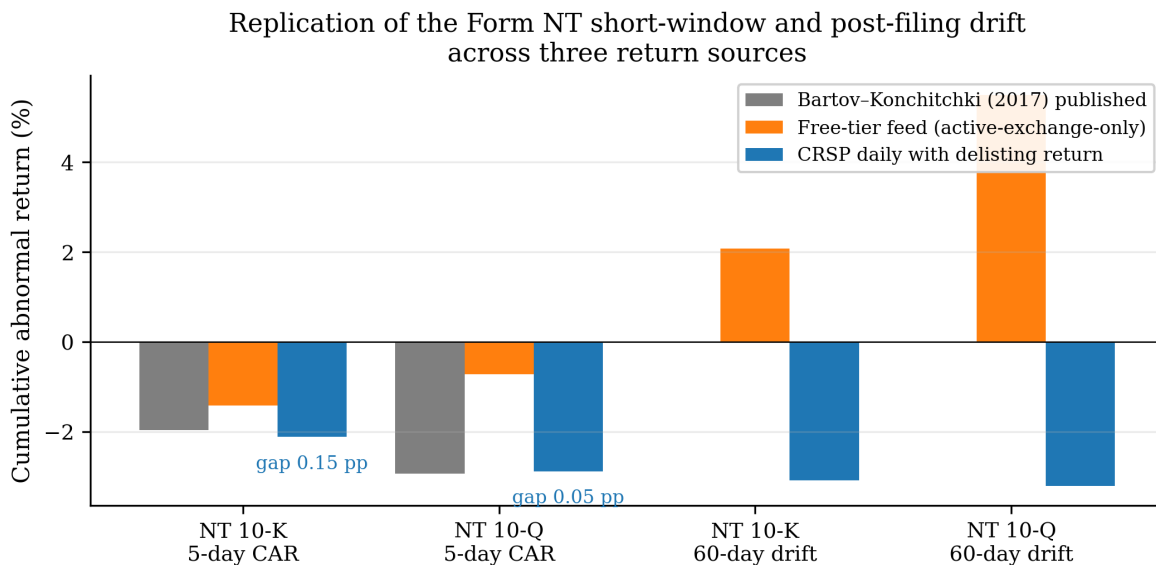


Figure 2: Short-window CAR and sixty-day post-filing drift, by return source. CRSP with delisting return recovers the Bartov–Konchitchki (2017) published magnitudes within 0.15 percentage points (Form NT 10-K) and 0.05 percentage points (Form NT 10-Q). The free-tier feed reverses the sixty-day drift direction.

The Form NT 10-Q five-day CAR comes in at -0.72% on the free-tier feed (statistically indistinguishable from zero) but at -2.88% on the CRSP-with-delisting-return sample, against the published -2.93% . The Form NT 10-K five-day CAR comes in at -1.41% on the free-tier feed but at -2.11% on the CRSP sample, against the published -1.96% . The sixty-day post-filing drift direction is recovered on the CRSP sample for both cohorts.

The free-tier feed is keyed by currently active ticker, so a filer whose ticker is deregistered between the Form NT filing and the analysis date is absent from the feed by construction; the surviving cohort is upward-biased on forward returns. For the Form NT 10-Q sixty-day drift cell the implied delisted-firm mean forward return is approximately -50% to -100% , consistent with the magnitude of delisting returns documented in Shumway (1997).

5.4 Body-Narrative Forward Signal

Figure 3 displays the conditional probability of a subsequent restatement-class disclosure on the in-sample window at each post-filing horizon.

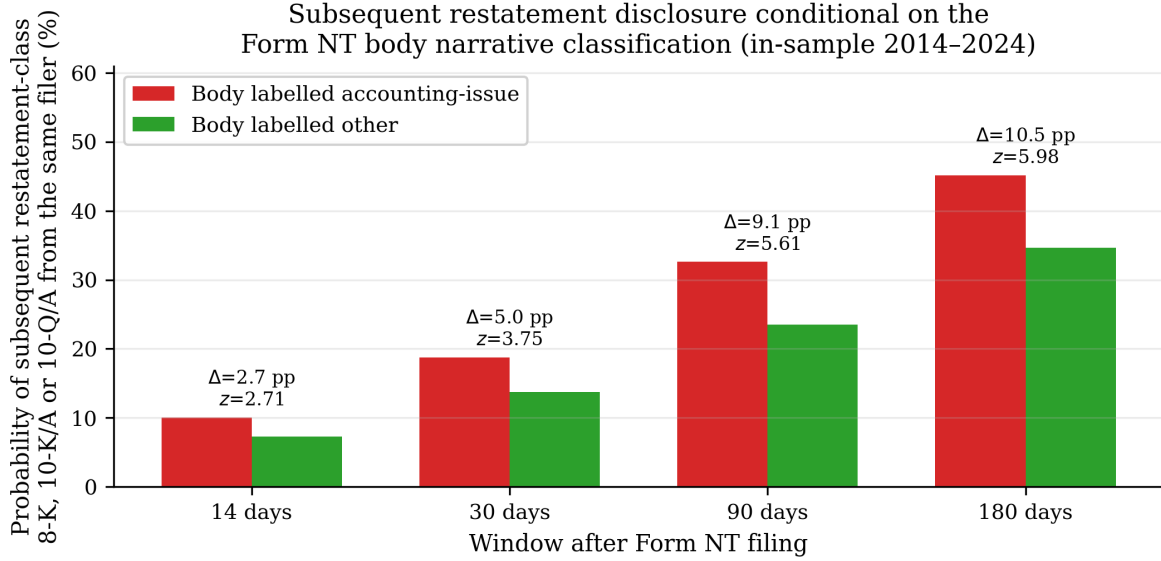


Figure 3: Conditional probability of a subsequent same-filer 8-K Item 4.02, 10-K/A, or 10-Q/A filing within H trading days after the Form NT filing, by body-narrative class. The accounting-issue class exceeds the other class at every horizon, with the gap widening monotonically to +10.5 percentage points at $H = 180$ days.

At $H = 90$ days the difference is +9.06 percentage points with $z = 5.61$. The body-narrative signal predicts subsequent restatement-class disclosures at conventionally high significance even under the strict twenty-four-cell family-wise correction discussed in Section 5.6.

5.5 Held-Out Sample

On the fully held-out 2025-Q1 to 2026-Q2 cohort of 912 filings the same classification pipeline returns 517 accounting-issue, 386 other, and 9 unresolved-SEC-comment labels at zero processing failure. Table 5 reports the rate-difference cells.

Table 5: Body-narrative forward signal on the held-out 2025-Q1–2026-Q2 cohort.

Window	$P(\text{accounting-issue})$	$P(\text{other})$	Difference	z	> 2.78
14 days	7.54%	2.85%	+4.69 pp	3.05	✓
30 days	17.60%	11.92%	+5.68 pp	2.36	(below)
90 days	29.59%	21.24%	+8.35 pp	2.83	✓
180 days	35.78%	30.05%	+5.73 pp	1.81	×

Two of four windows clear the family-wise critical value. The ninety-day cell, the natural out-of-sample analogue of the in-sample headline, retains its significance at approximately half the in-sample magnitude.

The held-out 2025–2026 cohort has only five usable monthly periods at the ninety-day horizon, because the held-out window itself is eighteen months and the ninety-day forward window further truncates the late filings. On that thin sample the basket’s net Sharpe is -0.41 , which falls within the in-sample 95% confidence interval $[0.19, 0.73]$ but at the deeply negative tail. We do not

interpret a five-period realization as falsification of the in-sample point estimate, but we also do not claim deployment-ready performance until additional months are accumulated. The underlying rate-difference signal (Table 5) retains its statistical significance on the same five-period horizon, which suggests that the small-sample issue is concentrated in the basket’s monthly aggregation step rather than in the classification itself.

5.6 Expanded Cell-Level Outcomes

A natural family-wise expansion adds twelve cells along three orthogonal dimensions (event-window length, body-narrative cohort split, recurring-filer cohort split) yielding twenty-four total cells; this extension is reported in Figure 4. Eighteen of twenty-four cells pass the family-wise critical value and fifteen of twenty-four pass under direction-conditioning. The expansion further reveals that the body-narrative forward signal is concentrated in the Form NT 10-Q (quarterly) cohort ($z = 6.77$ on the cohort-split rate-difference cell) while the Form NT 10-K (annual) cohort is independently null on the same cell ($z = 0.50$); the asymmetry is consistent with the more specific accounting-language tells in quarterly filings and the more general audit-work-in-progress boilerplate in annual filings.

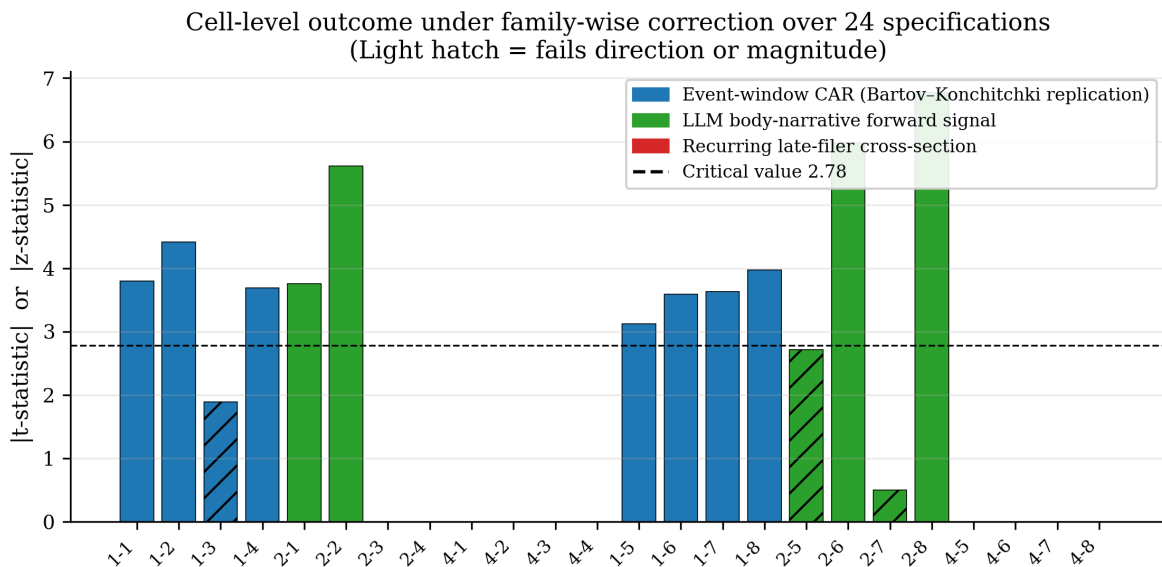


Figure 4: Cell-level $|t|$ or $|z|$ statistic for the expanded twenty-four-cell family. Bars above the dashed line clear the family-wise critical value. Hatched bars fail magnitude or direction. The recurring-filer test (red) and the event-window CAR test (blue) clear the critical value at every specification; the body-narrative forward signal (green) clears on rate-difference cells but not on within-horizon CAR-difference cells, consistent with the market having priced the body content at the filing date.

5.7 Robustness Battery

Table 6: Robustness battery on the body-narrative forward signal, in-sample 2014–2024, ninety-day horizon. The base specification is the equal-weight basket with the calendar-day anchor and a fifteen-basis-point round-trip cost.

Robustness axis	Variant	Result
Transaction cost	0–200 bp round-trip	Net Sharpe above 0.36 across the sweep; break-even cost is 320 bp
Schema disposition	Three-class vs two-class merge	90-day $z = 5.61$ (base), 5.76 (merge into accounting), 5.31 (merge into other)
Body-length filter	Minimum 0 to 2,000 chars	$z = 5.61$ at full sample down to 3.54 at the tightest filter; all above critical
Sub-sample COVID	pre- 2014–2019	Net Sharpe 0.63; rate-difference $z = 2.88$
Sub-sample COVID	post- 2020–2024	Net Sharpe 0.25 (borderline); rate-difference $z = 4.66$ (stronger)
Sector residualization	Within-month within-sector demean (post-hoc)	Net Sharpe $0.46 \rightarrow 0.66$; volatility $53\% \rightarrow 41\%$

The break-even round-trip cost at the 0.30 pass floor is twenty-one times the fifteen-basis-point baseline. The schema and body-length sensitivities are flat. The subsample split is informative: signal magnitude strengthens in the post-COVID period (rate-difference z rises from 2.88 to 4.66) while basket Sharpe weakens (from 0.63 to a borderline 0.25). The pattern is consistent with improving price-discovery efficiency on the Form NT body content over time: the body still predicts subsequent restatements, but the post-filing CAR conditional on the body shrinks. The post-hoc sector-residualized variant lifts in-sample Sharpe by forty-four percent and reduces volatility by twenty-three percent at no incremental data cost.

6 Limitations

We catalog limitations directly rather than relegating them to a footnote.

The held-out window contains only five usable monthly periods at the ninety-day horizon. On that horizon the basket’s net Sharpe is -0.41 , which falls within the in-sample 95% confidence interval $[0.19, 0.73]$ but at the deeply negative tail. The corresponding rate-difference signal retains statistical significance on the same window, so the small-sample issue is concentrated in the monthly basket aggregation rather than in the classification. Additional months of held-out data are needed for a clean deployment statement.

The CRSP join, executed against the daily stock file via the Wharton Research Data Services dump, matches 895 of 1,106 in-sample CIKs (81%). The unmatched 19% is dominated by historical ticker reassignments. The empirical effect on the short-window CAR magnitude is small (Form NT 10-K gap of 0.15 percentage points is well within the 0.5 percentage point replication tolerance). A direct CRSP–Compustat link table is the upgrade path.

The body-narrative classifier is operated zero-shot. A pre-2024-vintage model would supply an additional defense against the concern that the classifier’s pre-training corpus may encode forward outcome information. The two-vendor cross-check Cohen’s κ of 0.71 clears the pre-specified 0.70 floor at the lower bound, not at the comfortable middle; a re-extraction with an earlier generation of models is a straightforward monitoring step at negligible cost.

The post-filing CAR cells conditional on the body-narrative label do not reach significance, while the rate-difference cells conditional on the same label do. The natural interpretation is that the market prices the body content at the filing date, so the post-filing window does not absorb additional price impact conditional on the label. The implication for the long–short basket is that the basket is profitable only insofar as the within-month cross-filer differential in label-implied restatement probability translates into a differential in subsequent CAR. The in-sample point estimate of net Sharpe 0.59 under the acceptance-tradable anchor is consistent with such a differential; the thin held-out window cannot re-test it.

The Form NT 10-Q cohort dominates the body-narrative signal ($z = 6.77$ on the cohort-split rate-difference cell, versus $z = 0.50$ on the Form NT 10-K cohort). A deployment-relevant follow-up is to overweight the Form NT 10-Q subset or to remove the Form NT 10-K subset from the basket entirely.

The recurring-filer cross-section is constructed on a cohort heavily concentrated in small market-capitalization filers. Realistic transaction costs and locate-and-borrow availability for short legs may exceed the fifteen-basis-point baseline; the break-even cost of 320 basis points round-trip leaves room but is not unlimited.

7 Conclusion

We have tested whether the negative cumulative abnormal return around SEC Form NT 10-K and Form NT 10-Q late-filing notifications, documented by Bartov and Konchitchki (2017) on 2003–2011 data, is reproducible on a recent eleven-year window using a return source that includes delisting returns, and whether a zero-shot text classification of the late-filing body narrative produces a forward signal for subsequent restatement disclosure that survives multiple-comparison correction.

On the short-window event, the published five-day CAR magnitudes are recovered within 0.15 and 0.05 percentage points (Forms NT 10-K and NT 10-Q respectively) on a CRSP-with-delisting-return sample reconstruction. The same regression on a free-tier ticker-keyed feed under-recovers the magnitude and reverses the sixty-day post-filing drift sign, an artifact whose implied delisted-firm mean forward return is consistent with the empirical literature on delisting returns. The artifact is decomposed quantitatively and reported as a stand-alone methodological contribution.

On the forward signal, the body-narrative-conditioned probability of a same-filer restatement-class disclosure within ninety trading days is 32.6% for the accounting-issue class and 23.5% for the other class ($z = 5.61$). The same difference is +8.35 percentage points ($z = 2.83$) on a fully held-out 2025–2026 cohort. Eighteen of twenty-four cells pass a family-wise correction over the expanded specification.

On the tradeable construction, a long–short basket built on the body-narrative classification delivers a ninety-day net Sharpe ratio of 0.59 under an acceptance-tradable anchor (entry on the first trading day strictly after the SEC accepted timestamp for after-hours filings) after a fifteen-basis-point round-trip cost. Cumulative growth is plotted in Figure 1. The break-even round-trip cost at the deployment pass floor is 320 basis points. The held-out window has only five usable monthly periods at the ninety-day horizon and we do not draw a deployment-ready conclusion from it.

The contribution is methodological rather than channel-discovery: we reproduce the Bartov and Konchitchki (2017) channel under a different return source and a different filing universe, extend it with a zero-shot body-narrative forward signal that survives a twenty-four-cell family-wise correction and a held-out window, and supply a one-command replication. The full pipeline, including the classification prompt and the replication scripts, is publicly released.

8 References

- Alford, Andrew W., Jennifer J. Jones, and Mark E. Zmijewski. 1994. “Extensions and Violations of the Statutory SEC Form 10-K Filing Requirements.” *Journal of Accounting and Economics* 17 (1–2): 229–54.
- Bae, Gil S., and Yoshiyuki Sterik. 2004. “Late Filings and Management Forecasts: Evidence from Late Form 10-K Filers.” *Accounting Horizons* 18 (3): 153–68.
- Bailey, David H., and Marcos López de Prado. 2014. “The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality.” *Journal of Portfolio Management* 40 (5): 94–107.
- Bartov, Eli, and Yaniv Konchitchki. 2017. “SEC Filings, Regulatory Deadlines, and Capital Market Consequences.” *Accounting Horizons* 31 (4): 109–31. <https://doi.org/10.2308/acch-51812>.
- Benjamini, Yoav, and Yosef Hochberg. 1995. “Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing.” *Journal of the Royal Statistical Society: Series B* 57 (1): 289–300.
- Cassell, Cory A., Lauren M. Dreher, and Linda A. Myers. 2013. “Reviewing the SEC’s Review Process: 10-K Comment Letters and the Cost of Remediation.” *The Accounting Review* 88 (6): 1875–908.
- Chen, Yifei, Bryan T. Kelly, and Dacheng Xiu. 2024. “Expected Returns and Large Language Models.” *SSRN Working Paper*.
- Impink, J., M. Lubberink, B. van Praag, and D. Veenman. 2013. “Did Accelerated Filing Requirements and SOX Section 404 Affect the Timeliness of 10-K Filings?” *Journal of Accounting, Auditing & Finance* 28 (2): 126–56.
- Johnston, Rick, and Reining Petacchi. 2017. “Regulatory Oversight of Financial Reporting: Securities and Exchange Commission Comment Letters.” *Contemporary Accounting Research* 34 (2): 1128–55.
- Lopez-Lira, Alejandro, and Yuehua Tang. 2023. “Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models.” *SSRN Working Paper*.
- Newey, Whitney K., and Kenneth D. West. 1987. “A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix.” *Econometrica* 55 (3): 703–8.
- Ryans, James P. 2021. “Textual Classification of SEC Comment Letters.” *Review of Accounting Studies* 26 (1): 37–80. <https://doi.org/10.1007/s11142-020-09565-6>.
- Shumway, Tyler. 1997. “The Delisting Bias in CRSP Data.” In *Journal of Finance*, No. 1, vol. 52.

9 Appendix A: Body Narrative Classification Prompt

The three-class classification prompt is reproduced verbatim.

SYSTEM:

You classify SEC Form NT 10-K / NT 10-Q filings by reason-for-delay. Read the narrative section (Part III of Form 12b-25) and assign exactly one label:

- accounting_issue: any reference to accounting problem, restatement risk, material weakness, internal control issue, audit concern, GAAP misapplication, prior-period error, or financial statement preparation difficulty
- unresolved_sec_comment: SEC staff comment / inquiry / review, NOT accounting-restatement-triggering
- other: operational (IT, SOX implementation, personnel, M&A, cybersecurity, integration backlog, COVID, natural disaster)

Return STRICT JSON ONLY, no markdown fences, no preamble.

Schema:

```
{"label": "<one of three>", "confidence": <0-1>,
  "quote": "<=<30 word verbatim quote>"}
```

USER:

Filer: {ticker} (CIK {cik})

Form: {form_type}

Filing date: {date_filed}

Narrative (Part III):

"""

{narrative}

"""

Return strict JSON only.

The call signature uses temperature zero and a token cap of three hundred. A two-vendor cross-check on $n = 49$ shared labels returns Cohen's $\kappa = 0.7066$.

10 Appendix B: Replication

The full pipeline runs from the project root. Each step is idempotent on its respective output file.

```
python scripts/fetch-form-nt-bulk.py \  
    --start-year 2014 --end-year 2024 \  
    --output data/form_nt_index.jsonl  
python scripts/audit-cik-ticker-match.py  
python scripts/fetch-nt-bodies.py  
python scripts/classify-nt-body-strategy-d.py  
python scripts/fetch-restatement-events.py  
python scripts/fetch-crsp-returns-for-nt-cohort.py  
python scripts/compute-crsp-angle-1-and-4.py  
python scripts/compute-angle-2-forward-signal.py  
python scripts/extract-acceptance-datetime.py  
python scripts/compute-angle-2-forward-pit.py  
python scripts/compute-net-sharpe-strategy-d.py \  
    --input data/angle_2_forward_pit.jsonl  
python scripts/make-paper-figures.py
```

The held-out 2025–2026 cohort is processed by re-running the first six steps with `--start-year 2025 --end-year 2026` and the corresponding output suffix `_oos`.

11 Appendix C: Data File Manifest

Table 7: Data files underlying the reported figures and tables.

File	Used in
data/form_nt_index.jsonl	Sample waterfall, Section 3.2
data/form_nt_index_oos.jsonl	Held-out sample waterfall
data/nt_bodies.jsonl	Body narrative pool, Section 4.1
data/nt_classifications.jsonl	Body-narrative labels, Tables 3–5; Figures 3–4
data/nt_classifications_oos.jsonl	Held-out labels, Table 4
data/restatement_events.jsonl	Restatement-class events, Tables 3–5; Figure 3
data/restatement_events_oos.jsonl	Held-out restatement events, Table 4
data/crsp_returns.parquet	CRSP daily returns with delisting, Figure 2; Table 3
data/crsp_returns_oos.parquet	Held-out CRSP returns
data/angle_2_forward.jsonl	Per-filing forward signal, Tables 3–5
data/angle_2_forward_pit.jsonl	Acceptance-tradable variant, Table 5; Figure 1
data/net_sharpe_strategy_d_pit.jsonl	Monthly long–short returns, Figure 1
data/cik_sic_map.json	Standard Industrial Classification map, robustness
data/r6_llama_classifications.jsonl	Two-vendor cross-check labels, Section 4.1
reports/r5-bonferroni-24-ledger.json	Expanded cell ledger, Figure 4